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OSS Practice 1-2

Task 1:

View all processes

ps aux

```
Terminal
File  Edit  View  Search  Terminal  Help
[analyst@secOps ~]$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.5  0.6 167000 12424 ?        Ss   09:25   0:00 /sbin/init sp
root           2  0.0  0.0      0     0 ?        S    09:25   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        I<   09:25   0:00 [rcu_gp]
root           4  0.0  0.0      0     0 ?        I<   09:25   0:00 [rcu_par_gp]
root           5  0.0  0.0      0     0 ?        I<   09:25   0:00 [slub_flushwq
root           6  0.0  0.0      0     0 ?        I<   09:25   0:00 [netns]
root         891  3.9  6.3 415308 128272 tty7    Ssl+ 09:25   0:07 /usr/lib/xorg
```

the process with the highest memory consumption: 891 root

top

```
[analyst@secOps ~]$ top

top - 09:31:55 up 6 min,  1 user,  load average: 0.06, 0.12, 0.07
Tasks: 148 total,   1 running, 147 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.2 us,  0.2 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   1975.8 total,   1272.6 free,    354.7 used,    348.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   1450.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    891 root      20   0  419192 132060  68376 S   0.7   6.5   0:12.20 Xorg
    874 root      20   0  358660   2840   2472 S   0.3   0.1   0:00.10 VBoxService
   1305 analyst   20   0  216256   2408   2048 S   0.3   0.1   0:01.29 VBoxClient
   1523 analyst   20   0  339816  28144  21132 S   0.3   1.4   0:01.28 mate-multiload-
   1804 analyst   20   0   10488   3788   3200 R   0.3   0.2   0:00.08 top
      1 root      20   0  167000  12424   8192 S   0.0   0.6   0:01.01 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
```

the process with the highest memory consumption: 891 root

htop

```
  0[|                                    0.7%]   Tasks: 66, 114 thr; 2 running
  1[|                                    0.7%]   Load average: 0.17 0.14 0.09
Mem[||||||||||||||||                 385M/1.93G]   Uptime: 00:08:05
Swp[                                   0K/2.00G]

    PID USER      PRI  NI  VIRT   RES   SHR S CPU%▽MEM%   TIME+  Command (merged)
    891 root       20   0  413M  133M 68376 R  2.0  6.7  0:14.38 /usr/lib/xorg/Xorg -core :0 -seat sea
    899 root       20   0  413M  133M 68376 S  0.7  6.7  0:02.81 InputThread│/usr/lib/xorg/Xorg -core
   1305 analyst    20   0  211M  2408  2048 S  0.7  0.1  0:01.74 /opt/VBoxGuestAdditions-7.0.6/bin/VBo
   1318 analyst    20   0  211M  2408  2048 S  0.7  0.1  0:01.74 /opt/VBoxGuestAdditions-7.0.6/bin/VBo
   1427 analyst    20   0  983M 38312 29192 S  0.7  1.9  0:00.39 /usr/bin/mate-settings-daemon
   1515 analyst    20   0 1384M 65608 53348 S  0.7  3.2  0:01.33 /usr/bin/caja
   1523 analyst    20   0  331M 28144 21132 S  0.7  1.4  0:01.66 /usr/lib/mate-applets/mate-multiload-
      1 root       20   0  163M 12424  8192 S  0.0  0.6  0:01.01 systemd│/sbin/init splash
    253 root       19  -1 47872 15196 14028 S  0.0  0.8  0:00.18 /lib/systemd/systemd-journald
```

the process with the highest memory consumption: 891 root

Find parent process (PPID) and the path to the executable file

```
[analyst@secOps ~]$ ps -o pid,ppid,cmd -p 891
    PID    PPID CMD
    891     857 /usr/lib/xorg/Xorg -core :0 -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp v
```

Using process hierarchy tools, the process with the highest memory consumption was identified as **Xorg**, with a PID of **891**. The parent process (PPID) was **857**, and the executable path was **/usr/lib/xorg/Xorg**.

Based on analysis and comparison with the “Attackers and Their Tools” concepts, this process is a **legitimate system process**. Xorg is responsible for graphical display management in Linux, and it is executed from a standard system directory, which indicates normal behavior rather than a masked binary.

Task 2:

Find the authentication process

```
[analyst@secOps ~]$ ps aux | grep sshd
root         941  0.0  0.4  15424  8700 ?        Ss   09:25   0:00 sshd: /usr/sbin/sshd -D [listener]
0 of 10-100 startups
analyst     1890  0.0  0.1   6476  2304 pts/0    S+   09:48   0:00 grep sshd
```

Find configuration files in /etc/

```
[analyst@secOps ~]$ /etc/ssh/sshd_config
bash: /etc/ssh/sshd_config: Permission denied
[analyst@secOps ~]$ /etc/passwd
bash: /etc/passwd: Permission denied
[analyst@secOps ~]$ /etc/shadow
bash: /etc/shadow: Permission denied
[analyst@secOps ~]$ /etc/group
bash: /etc/group: Permission denied
[analyst@secOps ~]$ /etc/pam.d/
bash: /etc/pam.d/: Is a directory
```

The primary service responsible for user authentication in Linux is sshd, which manages secure user logins. Its main configuration file is located at /etc/ssh/sshd_config, while authentication-related data and rules are stored in /etc/passwd, /etc/shadow, /etc/group, and /etc/pam.d/.

Access to these files was denied when attempted as a standard user, demonstrating proper permission enforcement. Attackers frequently target these services because they control authentication and operate with elevated privileges. By modifying these files, an attacker can establish persistence and maintain long-term access to the system, as described in the “Attacking the Foundation” module.

Task 3:

First terminal: **ps -ef --forest**

Second terminal: **ping 8.8.8.8**

```
Terminal
File  Edit  View  Search  Terminal  Help
root         857       1  0 09:25 ?        00:00:00 /usr/sbin/lightdm
root         891     857  1 09:25 tty7     00:00:40  \_ /usr/lib/xorg/Xorg -core :0 -seat seat0
root        1143     857  0 09:26 ?        00:00:00  \_ lightdm --session-child 12 19
analyst     1161    1143  0 09:26 ?        00:00:00      \_ mate-session
analyst     1427    1161  0 09:26 ?        00:00:00          \_ /usr/bin/mate-settings-daemon
analyst     1465    1161  0 09:26 ?        00:00:03          \_ marco
analyst     1483    1161  0 09:26 ?        00:00:00          \_ mate-panel
analyst     1515    1161  0 09:26 ?        00:00:02          \_ /usr/bin/caja
analyst     1693    1515  0 09:27 ?        00:00:08          |   \_ mate-terminal
analyst     1938    1693  0 10:00 pts/1    00:00:00          |       \_ bash
analyst     1947    1938  0 10:01 pts/1    00:00:00          |       |   \_ ping 8.8.8.8
analyst     1945    1693  0 10:00 pts/0    00:00:00          |       \_ bash
analyst     1954    1945  0 10:06 pts/0    00:00:00          |           \_ ps -ef --forest
analyst     1535    1161  0 09:26 ?        00:00:00          \_ mate-power-manager
analyst     1564    1161  0 09:26 ?        00:00:00          \_ /usr/lib/x86_64-linux-gnu/polkit
root         871       1  0 09:25 ?        00:00:01 /usr/bin/VBoxDRMClient
```

```
Terminal
File  Edit  View  Search  Terminal  Help
[analyst@secOps ~]$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=56.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=55.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=56.0 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=58.3 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=56.3 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=255 time=55.7 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=255 time=56.7 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=255 time=56.3 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=255 time=57.4 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=255 time=59.2 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=255 time=56.4 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=255 time=56.7 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=255 time=57.8 ms
64 bytes from 8.8.8.8: icmp_seq=14 ttl=255 time=57.7 ms
```

A network-related tool (ping) was launched from a terminal while observing the process tree in real time. The process appeared as a child of the shell and terminal process. When the parent terminal was closed, the child process terminated automatically.

Malware can exploit this behavior by detaching itself from the parent process and becoming an orphan process. Once adopted by init or systemd, the malicious process can continue running silently in the background, allowing it to remain hidden and maintain persistence.

Task 4:

Create new User

```
[analyst@secOps ~]$ sudo useradd Test_User
[sudo] password for analyst:
Sorry, try again.
[sudo] password for analyst:
[analyst@secOps ~]$ sudo passwd Test_User
New password:
Retype new password:
passwd: password updated successfully
```

Log in as a new user

```
[analyst@secOps ~]$ su - Test_User
Password:
su: warning: cannot change directory to /home/Test_User: No such file or directo
ry
-sh: 1: /etc/profile.d/no-screensaver.sh: [[: not found
$
```

Attempt to access protected resources

```
$ cd root
-sh: 1: cd: can't cd to root
$ cat /var/log/auth.log
cat: /var/log/auth.log: Permission denied
```

Temporary provision of sudo access

```
[analyst@secOps ~]$ sudo usermod -aG sudo Test_User
[sudo] password for analyst:
```

A standard user named Test_User was created and attempts to access protected directories such as /root and sensitive log files in /var/log resulted in “Permission denied,” confirming proper privilege separation.

The user was then temporarily granted administrative privileges using sudo. Leaving a standard user in the sudoers group poses a significant security risk, as a compromised account would allow an attacker to gain full control of the system, modify critical files, and establish persistence.

Task 5:

Viewing active network connections

```
[analyst@secOps ~]$ sudo ss -tulnp
Netid  State   Recv-Q   Send-Q                          Local Address:Port          Peer Address:Port       Process
udp    UNCONN  0        0                                     0.0.0.0:54041              0.0.0.0:*           users:(("firefox",pid=2057,fd=159))
udp    UNCONN  0        0                                     0.0.0.0:46920              0.0.0.0:*           users:(("firefox",pid=2057,fd=91))
udp    UNCONN  0        0                                     0.0.0.0:55203              0.0.0.0:*           users:(("firefox",pid=2057,fd=154))
udp    UNCONN  0        0                                 127.0.0.53%lo:53               0.0.0.0:*           users:(("systemd-resolve",pid=383,fd=13))
udp    UNCONN  0        0                               10.0.2.15%enp0s3:68              0.0.0.0:*           users:(("systemd-network",pid=345,fd=17))
udp    UNCONN  0        0                                     0.0.0.0:40040              0.0.0.0:*           users:(("firefox",pid=2057,fd=111))
udp    UNCONN  0        0                                    10.0.2.15:123               0.0.0.0:*           users:(("ntpd",pid=687,fd=23))
udp    UNCONN  0        0                                    127.0.0.1:123               0.0.0.0:*           users:(("ntpd",pid=687,fd=18))
udp    UNCONN  0        0                                      0.0.0.0:123               0.0.0.0:*           users:(("ntpd",pid=687,fd=17))
udp    UNCONN  0        0                                      0.0.0.0:56000             0.0.0.0:*           users:(("firefox",pid=2057,fd=165))
udp    UNCONN  0        0          [fd17:625c:f037:2:a00:27ff:fe55:4407]:123                [::]:*           users:(("ntpd",pid=687,fd=25))
udp    UNCONN  0        0              [fe80::a00:27ff:fe55:4407]%enp0s3:123                [::]:*           users:(("ntpd",pid=687,fd=24))
udp    UNCONN  0        0                                          [::1]:123                [::]:*           users:(("ntpd",pid=687,fd=19))
udp    UNCONN  0        0                                           [::]:123                [::]:*           users:(("ntpd",pid=687,fd=16))
tcp    LISTEN  0        32                                     0.0.0.0:21                 0.0.0.0:*           users:(("vsftpd",pid=669,fd=3))
tcp    LISTEN  0        4096                              127.0.0.53%lo:53               0.0.0.0:*           users:(("systemd-resolve",pid=383,fd=14))
tcp    LISTEN  0        128                                    0.0.0.0:22                 0.0.0.0:*           users:(("sshd",pid=941,fd=3))
tcp    LISTEN  0        128                                       [::]:22                    [::]:*           users:(("sshd",pid=941,fd=4))
tcp    LISTEN  0        64                                           *:23                       *:*           users:(("xinetd",pid=950,fd=5))
```

Correlation with Task 1

Process: firefox
The PID matches
Launched from a standard user environment
Not from /tmp, /dev/shm, /var/tmp
Full correlation with the processes from Task 1, there are no signs of masking.

Is there anything suspicious?

No.

Why
there are no non-standard listening ports
no unknown binaries
no external IP addresses associated with background daemons
the activity is explained by user actions

Active network connections were analyzed using the ss command. In addition to system services, the firefox process was observed using multiple high-numbered UDP ports. These ports represent ephemeral connections initiated by the web browser for normal network communication such as web traffic and DNS resolution.

The firefox process was correlated with the process list from Task 1 and was confirmed to be a legitimate user application running from standard directories. No processes were found communicating externally in a suspicious manner or listening on unusual ports, and no indicators of malicious network activity were identified.

Task 6:

Find the SSH service configuration

```
[analyst@secOps ~]$ /etc/ssh/sshd_config
bash: /etc/ssh/sshd_config: Permission denied
[analyst@secOps ~]$ sudo nano /etc/ssh/sshd_config
```

```
  GNU nano 6.2                    /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/us>

# The strategy used for options in the default sshd_config shipped with
```

Change the parameter

```
#PermitRootLogin prohibit-password
#PermitRootLogin no
```

This prohibits direct root login via SSH.
Save the file and log out.

Restart the service

```
[analyst@secOps ~]$ sudo systemctl restart ssh
[analyst@secOps ~]$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2025-12-23 10:37:17 UTC; 13s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 2468 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 2469 (sshd)
      Tasks: 1 (limit: 2287)
     Memory: 1.7M
        CPU: 29ms
     CGroup: /system.slice/ssh.service
             └─2469 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 23 10:37:17 labvm systemd[1]: Starting OpenBSD Secure Shell server...
Dec 23 10:37:17 labvm sshd[2469]: Server listening on 0.0.0.0 port 22.
Dec 23 10:37:17 labvm sshd[2469]: Server listening on :: port 22.
Dec 23 10:37:17 labvm systemd[1]: Started OpenBSD Secure Shell server.
```

In Linux systems, configuration files located in /etc/ serve a role similar to the registry in other operating systems. The SSH service configuration file /etc/ssh/sshd_config was identified and modified to improve security.

The parameter PermitRootLogin was set to no, and the SSH service was restarted to apply the change. Improper configuration of services can create serious security vulnerabilities, as attackers often exploit weak or default settings rather than software flaws. This aligns with the risks discussed in Module 17 regarding insecure system configurations.

Task 7:

Launch the performance monitor

```
  0[|                          1.4%]   Tasks: 76, 278 thr; 2 running
  1[                           0.0%]   Load average: 0.46 0.21 0.13
Mem[||||||||||||||||||786M/1.93G]   Uptime: 01:15:35
Swp[                       0K/2.00G]

  PID USER      PRI  NI  VIRT   RES   SHR S CPU%▽MEM%   TIME+  Command
 2507 analyst    20   0  8432  4352  3480 R  1.3  0.2  0:00.03 htop
  891 root       20   0  438M  158M 76980 S  0.7  8.0  1:06.81 /usr/lib/xorg/X
 1297 analyst    20   0  210M  2528  2172 S  0.7  0.1  0:03.27 /usr/bin/VBoxCl
 1305 analyst    20   0  211M  2408  2048 S  0.7  0.1  0:18.36 /usr/bin/VBoxCl
 1318 analyst    20   0  211M  2408  2048 S  0.7  0.1  0:18.36 /usr/bin/VBoxCl
 1962 analyst    20   0  879M 54012 40764 S  0.7  2.7  0:04.44 mate-terminal
    1 root       20   0  163M 12424  8192 S  0.0  0.6  0:01.11 /sbin/init spla
  253 root       19  -1 47872 15268 14100 S  0.0  0.8  0:00.30 /lib/systemd/sy
  299 root       20   0 25964  6352  4572 S  0.0  0.3  0:00.21 /lib/systemd/sy
  345 systemd-n  20   0 16236  8172  7132 S  0.0  0.4  0:00.19 /lib/systemd/sy
```

Create a high load

```
[analyst@secOps ~]$ yes > /dev/null
```

Watch the splash

```
  0[||                          0.3%]   Tasks: 77, 278 thr; 2 running
  1[||||||||||||||||||||||||100.0%]   Load average: 0.96 0.35 0.18
Mem[||||||||||||||||||||786M/1.93G]   Uptime: 01:17:05
Swp[                        0K/2.00G]

  PID USER      PRI  NI  VIRT   RES   SHR S CPU%▽MEM%   TIME+  Command
 2509 analyst    20   0  5764  1020   928 R 199.  0.1  0:43.56 yes
  891 root       20   0  438M  158M 76980 R  0.4  8.0  1:07.38 /usr/lib/xorg/X
 1282 analyst    20   0  210M  2328  1788 S  0.2  0.1  0:00.13 /usr/bin/VBoxCl
 1568 analyst    20   0  476M 29076 21688 R  0.2  1.4  0:00.14 mate-power-mana
 1627 root       20   0  233M  8820  7672 S  0.2  0.4  0:00.17 /usr/libexec/up
 2057 analyst    20   0 3129M  368M  153M R  0.2 18.7  0:21.33 /usr/lib/firefo
 2075 analyst    20   0 3129M  368M  153M S  0.2 18.7  0:01.36 /usr/lib/firefo
 2507 analyst    20   0  8432  4692  3568 R  0.2  0.2  0:00.25 htop
    1 root       20   0  163M 12424  8192 S  0.0  0.6  0:01.11 /sbin/init spla
  253 root       19  -1 47872 15268 14100 S  0.0  0.8  0:00.30 /lib/systemd/sy
  299 root       20   0 25964  6352  4572 S  0.0  0.3  0:00.21 /lib/systemd/sy
```

System performance was monitored using htop while a high-load command was executed. A noticeable spike in CPU usage was observed, caused by a user-initiated process. Once the command was terminated, CPU usage immediately returned to normal levels.

This behavior indicates legitimate activity rather than a Denial of Service attack or a cryptominer. Malicious processes typically maintain sustained high resource usage, often restart automatically, and may include external network communication, none of which were observed in this case.

Task 8:

Go to the logs directory

```
[analyst@secOps ~]$ cd /var/log
[analyst@secOps log]$ ls
alternatives.log       dpkg.log               ntpstats
apt                    faillog                openvswitch
auth.log               fontconfig.log         private
bootstrap.log          freeradius             snort
btmp                   gpu-manager.log        syslog
cloud-init.log         installer              ubuntu-advantage.log
cloud-init-output.log  journal                ubuntu-advantage-timer.log
dist-upgrade           kern.log               unattended-upgrades
dmesg                  landscape              vboxadd-install.log
dmesg.0                lastlog                vboxadd-setup.log
dmesg.1.gz             lightdm                wtmp
dmesg.2.gz             logstash-tutorial.log  Xorg.0.log
dmesg.3.gz             nginx                  Xorg.0.log.old
dmesg.4.gz             nginx-logstash.log
```

Auth.log analysis

```
[analyst@secOps log]$ sudo cat auth.log
[sudo] password for analyst:
Feb 10 21:10:55 labvm useradd[799]: new group: name=setup, GID=1000
Feb 10 21:10:55 labvm useradd[799]: new user: name=setup, UID=1000, GID=1000, ho
me=/home/setup, shell=/bin/bash, from=none
Feb 10 21:10:55 labvm useradd[799]: add 'setup' to group 'adm'
Feb 10 21:10:55 labvm useradd[799]: add 'setup' to group 'cdrom'
Feb 10 21:10:55 labvm useradd[799]: add 'setup' to group 'sudo'
Feb 10 21:10:55 labvm useradd[799]: add 'setup' to group 'dip'
Feb 10 21:10:55 labvm useradd[799]: add 'setup' to group 'plugdev'
Feb 10 21:10:55 labvm useradd[799]: add 'setup' to group 'lxd'
```

Syslog Analysis

```
[analyst@secOps log]$ sudo grep -i "error" syslog
Feb 10 21:10:55 labvm systemd[1]: Condition check resulted in Process error repo
rts when automatic reporting is enabled (file watch) being skipped.
Feb 10 21:10:55 labvm systemd[1]: Condition check resulted in Process error repo
rts when automatic reporting is enabled (timer based) being skipped.
Feb 10 21:10:55 labvm kernel: [    1.422889] RAS: Correctable Errors collector i
nitialized.
Feb 10 21:10:55 labvm kernel: [    2.371076] [drm:vmw_host_printf [vmwgfx]] *ERR
OR* Failed to send host log message.
```

System logs located in /var/log/ were analyzed, with a focus on auth.log and syslog. The authentication log revealed records of user authentication attempts and sudo usage, including timestamps, user accounts, and executed commands.

No evidence of repeated failed login attempts or unauthorized sudo usage was identified during the analysis period. These logs serve as a primary source for CyberOps investigations, as they provide a detailed timeline of system activity and are essential for detecting, analyzing, and responding to security incidents.

Task 9:

Based on the analysis of active network connections and system logs, an Indicator of Compromise was identified in the form of an active Telnet service listening on port 23. Telnet is considered insecure because it transmits data in plaintext and is commonly targeted by attackers for credential interception and unauthorized access.

This indicator is classified as malicious rather than merely suspicious because it represents a known security weakness that can be actively exploited, increasing the system's attack surface even in the absence of current exploitation evidence.

Task 10:

Final Incident Summary

Incident Overview:

A suspicious alert triggered an investigation on the CyberOps workstation. A deep-dive analysis of system processes, user privileges, network activity, configurations, performance metrics, and system logs was conducted to identify potential malicious behavior.

Findings:

Process analysis revealed that the highest memory-consuming process was a legitimate system process running from a standard system directory. Authentication services and configuration files were properly protected, and permission enforcement was confirmed. Network analysis showed only expected system and user-initiated connections, with no unauthorized external communication detected. System performance monitoring identified a temporary CPU spike caused by a user-initiated process, which was not indicative of a DoS attack or cryptominer activity. Log analysis of

authentication and system logs did not reveal repeated failed login attempts or unauthorized privilege escalation.

Indicator of Compromise (IOC):
An active Telnet service listening on port 23 was identified. Telnet is considered insecure due to plaintext communication and represents a known exploitable weakness, increasing the system's attack surface.

Impact Assessment:
No evidence of active exploitation or system compromise was found during the investigation. However, the presence of legacy services poses a potential security risk if left unmitigated.

Conclusion and Recommendations:
The system showed no signs of active malware or compromise at the time of analysis. It is recommended to disable insecure legacy services such as Telnet, continue monitoring logs and network activity, and maintain secure configuration practices to reduce future risk.